

AKILESHWARAKRISHNAN A

AWS Cloud Support & Backend Developer (Fresher) | AI & Data Science

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PROFESSIONAL SUMMARY

Certified AWS Solutions Architect and AI & Data Science graduate with hands-on experience in backend development (Python, REST APIs) and cloud architectures. Currently a Backend Developer Intern at Ashok Leyland, with a track record of building **cloud-hosted NoSQL applications**, deploying AI models, and optimizing routing algorithms. Passionate about Cloud Support, DevOps, and Backend Engineering.

EXPERIENCE

Backend Developer Intern | Ashok Leyland

July 2026 – Present | Chennai, Tamil Nadu

- Built and enhanced internal web applications using **Python** web frameworks to support day-to-day engineering operations.
- Designed and integrated secure, optimized **REST APIs** enabling reliable frontend-backend communication.
- Wrote automated unit tests with **Pytest** to raise code coverage and catch regression bugs before deployment.
- Restructured SQL queries and refactored server-side workflows, cutting application load time.
- Diagnosed and resolved performance bottlenecks through manual and automated testing cycles.

EDUCATION

- B.Tech (AI & Data Science): Nehru Institute of Engineering and Technology, Coimbatore | CGPA: 7.5 / 10 | 2022 – 2026
- HSC (Science): Srinivasa Vidhyalaya MHSS, Udumalpet | Score: 79.3% | May 2022

TECHNICAL SKILLS

- Languages: Python, Java, SQL
- Cloud & DevOps: AWS (EC2, S3, VPC, IAM, Load Balancers)
- Databases & Tools: Firebase Firestore (NoSQL), MySQL, Git, GitHub, VS Code
- AI/ML & Software Concepts: Q-Learning, DQN, NetworkX, NS-3, REST APIs, OOP, Responsive Design, IoT (ESP32)

PROJECTS

Sovereign Routing: AI-Driven Disaster Communication Network | 2026

Python, NetworkX, NS-3 Simulator, Q-Learning, DQN, ESP32, LoRa

- Engineered a distributed Q-Learning and DQN routing framework on ESP32 mesh nodes for backup communication during network outages.
- Achieved a **93% Packet Delivery Ratio**, **56% latency reduction**, and **66% faster recovery** under 40% node failures.
- Validated the framework via NetworkX/NS-3 simulations and a 5-node physical Wi-Fi/LoRa testbed.

Codebase: github.com/Akileshwarakrishnan/Sovereign-Routing-AI

Sri Sabari: Real-Time Appointment Booking System (Full-Stack) | 2026

Vanilla HTML5, CSS3, JavaScript (ES6+), Firebase Firestore & Auth, WhatsApp API, GitHub Pages

- Built a customer booking portal and a private administrative dashboard on a serverless architecture.
- Synced booking slots and dynamic pricing instantly using Firestore **onSnapshot** listeners.
- Managed the full order lifecycle (Pending, Confirmed, Completed) and integrated the WhatsApp API for automated alerts.

Codebases: github.com/Akileshwarakrishnan/Appointment-Booking-System (Client) | github.com/Akileshwarakrishnan/Appointment-Booking-System-admin (Admin)

Order & Inventory Tracking Application (Freelance) | 2026

Vanilla HTML5, CSS3, JS, Firebase Firestore, WhatsApp API

- Delivered a dual-portal inventory system for a commercial client, featuring secure authentication, cart workflows, and automated order alerts.

Codebase: github.com/Akileshwarakrishnan/greenlifefaturalfoods

CERTIFICATIONS

- AWS Certifications: AWS Certified Solutions Architect – Associate | AWS Certified Cloud Support Associate
- IT Certifications: Google Data Analytics | IBM Machine Learning | Python for Data Science | NPTEL IoT

RESEARCH PUBLICATIONS

Co-author: “Sovereign Routing: An AI-Driven Framework for Disaster-Resilient Communication Networks” — Department of AI & Data Science, NIET, Coimbatore (Published 2026).